Final Consulting Project Report

Tomato Crop Soil Health

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Contents

1 Summary

This project focused on evaluating soil health and crop response in a tomato field experiment designed to study the effects of cover cropping, Fusarium infection, and subplot treatments on soil aggregation, measured using WSA. The experimental design followed a split-plot structure, with cover crop and Fusarium infection defining the whole-plot factors and AMF, fungicide, and water assigned randomly as subplot treatments. The primary analysis used ANOVA for a split-plot design with random effects, along with targeted contrasts to address whether Fusarium-infected and non-infected plots differed and whether cover cropping improved soil health under the different treatment conditions.

A second component of the project examined whether crop color, measured through satellite-derived spectral data, could be used to indicate plant carbon and nitrogen levels. To support this objective, the spectral variables were separated into single-band and multi-band groups, and factor analysis was applied to each group to construct lower-dimensional latent factors for interpretation and potential regression modeling. Although data limitations prevented the full carbon and nitrogen modeling stage from being completed, the analysis established a framework for dimensionality reduction and demonstrated how latent-factor methods could be used for this type of agricultural monitoring problem.

2 Project Objectives

2.1 Problem Statement

The client sought to determine whether soil aggregation, measured by WSA, differed between Fusarium-infected and non-infected plots under cover-crop and no-cover-crop conditions, and whether subplot treatments consisting of AMF, fungicide, or water altered those outcomes. The central statistical problem was to evaluate these treatment and disease effects while properly accounting for the split-plot layout of the experiment.

A related problem involved understanding whether satellite-derived spectral measurements of crop color could serve as useful indicators of plant carbon and nitrogen levels. The challenge in this part of the project was that the spectral data were collinear and naturally grouped into single-band and multi-band measurements, requiring dimension-reduction methods before any meaningful regression analysis could be attempted. Taken together, the project addressed two connected questions: how management and disease conditions influence soil health in an agricultural experiment, and how remote sensing data might support future monitoring of crop nutrient status.

2.2 Project Objectives

The objectives of this project were:

- How is the soil aggregation affected by the joint effects of AMF and cover crops under tomato vegetation in both Fusarium infected and non-infected crops? How does soil aggregation affects soil physical properties like porosity and aeration?
- Does the amount of total C-N in soil aggregates differs to that of the total C-N in bulk soil? Does it correlate with the physical appearance of the tomato crops?

3 Deliverables

The project deliverables included a written report, reproducible analysis code, and supporting explanation of the statistical methods so that the client could understand, replicate, and extend the work. In addition to producing the formal analyses, the project emphasized clear communication of the split-plot modeling strategy, treatment contrasts, and factor-analytic methods in a way that supported practical decision-making and future use by the client.

4 Data

The field dataset came from a tomato crop experiment designed and conducted by the client to study the effects of soil management, disease pressure, and treatment application on soil health. The experimental layout followed a split-plot structure. At the whole-plot level, the field was divided into cover-crop and no-cover-crop conditions, and within each of these, plots were further divided into Fusarium-infected and non-infected sections. Within each resulting quarter, subplot treatments were randomly assigned as AMF, fungicide, or water, with water serving as the placebo treatment. The primary response variable for this component of the study was SA, which was used as the measure of soil aggregation.

The second dataset consisted of spectral measurements collected through satellite imagery, along with corresponding plant carbon and nitrogen measurements. These spectral variables were organized into two groups, single-band and multi-band data, to support separate exploratory analyses of their latent structure. This portion of the data was intended to evaluate whether remotely sensed crop color information could provide useful information about plant carbon and nitrogen status. Together, these datasets supported both the experimental soil-health analysis and the exploratory nutrient-monitoring component of the project.

5 Methods

5.1 Methods: Split-Plot Analysis of Soil Aggregation

The primary objective of this component of the project was to evaluate whether soil aggregation differed across combinations of cover-crop status, *Fusarium* infection status, and subplot treatment.

The response variable for this analysis was water-stable aggregates (WSA), which was used as the soil aggregation metric. The experiment followed a split-plot design, so the modeling strategy was chosen to reflect the hierarchical treatment assignment and the resulting dependence structure in the data.

5.1.1 Experimental Design

The field was first divided into whole plots based on two factors: cover-crop status and *Fusarium* infection status. This produced four whole-plot conditions:

$$\text{c-f}, \quad \text{c-nf}, \quad \text{nc-nf}, \quad \text{nc-f},$$

where

- c-f: cover crop with *Fusarium* infection,
- c-nf: cover crop without *Fusarium* infection,
- nc-nf: no cover crop without *Fusarium* infection,
- nc-f: no cover crop with *Fusarium* infection.

Within each whole plot, subplot treatments were assigned at random with three levels:

$$\text{W}, \quad \text{AMF}, \quad \text{F},$$

where W denotes water, AMF denotes arbuscular mycorrhizal fungi, and F denotes fungicide. Water served as the placebo treatment. Data were available from two years, 2024 and 2025, and year was included in the model as a nuisance factor.

5.1.2 Split-Plot Mixed Model

Let Y_{ijk} denote the WSA response for the kth observational unit in whole plot i and treatment level j. The split-plot mixed model used for analysis was

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_\ell + (\alpha\beta)_{ij} + u_{ik} + \varepsilon_{ijk},$$

where:

$$
\begin{aligned}
Y_{ijk} &= \text{WSA response for the } k\text{th unit under whole plot } i \text{ and treatment } j, \\
\mu &= \text{overall mean}, \\
\alpha_i &= \text{fixed effect of the } i\text{th whole-plot condition}, \\
\beta_j &= \text{fixed effect of the } j\text{th subplot treatment}, \\
\gamma_\ell &= \text{fixed effect of year } \ell, \\
(\alpha\beta)_{ij} &= \text{interaction effect between whole-plot condition and treatment}, \\
u_{ik} &= \text{random effect for split-plot unit nested within whole plot } i, \\
\varepsilon_{ijk} &= \text{residual error term.}
\end{aligned}
$$

The random terms were assumed to satisfy

$$u_{ik} \sim N(0, \sigma_u^2), \qquad \varepsilon_{ijk} \sim N(0, \sigma^2),$$

with independence between u_{ik} and ε_{ijk}.

In implementation, the model was fit as a linear mixed-effects model with fixed effects for whole plot, treatment, their interaction, and year, together with a random intercept for plot number:

$$\text{WSA} \sim \text{Whole\_Plot} * \text{Treatment} + \text{Year} + (1 \mid \text{Plot\_No}).$$

This specification was chosen to account for the split-plot structure of the experiment while allowing direct inference on whole-plot effects, treatment effects, and their interaction.

5.1.3 Cell-Mean Parameterization

For the post hoc contrast analysis, inference was expressed in terms of the cell means for the whole-plot by treatment combinations. Let

$$\mu = (\mu_{11}, \mu_{12}, \mu_{13}, \mu_{21}, \mu_{22}, \mu_{23}, \mu_{31}, \mu_{32}, \mu_{33}, \mu_{41}, \mu_{42}, \mu_{43})^T,$$

where μ_{ij} denotes the mean WSA value for whole-plot condition i and treatment j, with the indexing:

i	1	2	3	4
Whole plot	c-f	c-nf	nc-nf	nc-f

j	1	2	3
Treatment	W	AMF	F

If c is a contrast vector, then a linear contrast is written as $c^T\mu$, provided the contrast is estimable under the fitted model.

5.1.4 Planned Hypothesis Tests

The main scientific questions were addressed through planned contrasts comparing treatment effects across cover-crop conditions separately within *Fusarium*-infected and non-infected settings.

Contrasts under *Fusarium*-infected conditions. The first set of hypotheses examined whether the effect of AMF relative to the comparison treatments differed between cover-crop and no-cover-crop whole plots when *Fusarium* was present.

1. **AMF versus water under *Fusarium*:**

$$H_0 : \frac{\mu_{12} - \mu_{11}}{2} - \frac{\mu_{42} - \mu_{41}}{2} = 0 \qquad \text{vs.} \qquad H_A : \frac{\mu_{12} - \mu_{11}}{2} - \frac{\mu_{42} - \mu_{41}}{2} \neq 0.$$

2. **AMF versus fungicide under *Fusarium*:**

$$H_0 : \frac{\mu_{12} - \mu_{13}}{2} - \frac{\mu_{42} - \mu_{43}}{2} = 0 \qquad \text{vs.} \qquad H_A : \frac{\mu_{12} - \mu_{13}}{2} - \frac{\mu_{42} - \mu_{43}}{2} \neq 0.$$

3. **AMF versus the average of water and fungicide under *Fusarium*:**

$$H_0 : \frac{\mu_{12} - \frac{\mu_{11}+\mu_{13}}{2}}{2} - \frac{\mu_{42} - \frac{\mu_{41}+\mu_{43}}{2}}{2} = 0 \qquad \text{vs.} \qquad H_A : \frac{\mu_{12} - \frac{\mu_{11}+\mu_{13}}{2}}{2} - \frac{\mu_{42} - \frac{\mu_{41}+\mu_{43}}{2}}{2} \neq 0.$$

Contrasts under non-*Fusarium* conditions. The second set of hypotheses examined the same treatment comparisons under the non-infected conditions.

4. **AMF versus water under non-*Fusarium*:**

$$H_0 : \frac{\mu_{22} - \mu_{21}}{2} - \frac{\mu_{32} - \mu_{31}}{2} = 0 \qquad \text{vs.} \qquad H_A : \frac{\mu_{22} - \mu_{21}}{2} - \frac{\mu_{32} - \mu_{31}}{2} \neq 0.$$

5. **AMF versus fungicide under non-*Fusarium*:**

$$H_0 : \frac{\mu_{22} - \mu_{23}}{2} - \frac{\mu_{32} - \mu_{33}}{2} = 0 \qquad \text{vs.} \qquad H_A : \frac{\mu_{22} - \mu_{23}}{2} - \frac{\mu_{32} - \mu_{33}}{2} \neq 0.$$

6. **AMF versus the average of water and fungicide under non-*Fusarium*:**

$$H_0 : \frac{\mu_{22} - \frac{\mu_{21}+\mu_{23}}{2}}{2} - \frac{\mu_{32} - \frac{\mu_{31}+\mu_{33}}{2}}{2} = 0 \qquad \text{vs.} \qquad H_A : \frac{\mu_{22} - \frac{\mu_{21}+\mu_{23}}{2}}{2} - \frac{\mu_{32} - \frac{\mu_{31}+\mu_{33}}{2}}{2} \neq 0.$$

These planned contrasts were evaluated using estimated marginal means from the fitted mixed model. This allowed the comparisons of interest to be carried out directly on the fitted cell means while preserving the split-plot error structure of the experiment.

5.1.5 Modeling Rationale

The split-plot mixed-model framework was used because the treatment factors were applied at different experimental levels. Cover-crop status and *Fusarium* infection defined the whole-plot structure, while AMF, fungicide, and water were assigned within whole plots as subplot treatments. A mixed-effects model was therefore appropriate for accounting for the hierarchical nature of the experimental design and for ensuring that hypothesis tests and planned contrasts were conducted using an error structure aligned with the way the experiment was implemented.

5.2 Methods: Spectral Data and Carbon–Nitrogen Analysis

A second objective of the project was to investigate whether spectral measurements of crop color could be used to explain variation in plant carbon and nitrogen levels. The available spectral predictors were divided into two groups: single-band measurements and multiple-band measurements. Because each group contained several correlated variables, the analysis focused first on dimension reduction and latent-structure assessment before any regression modeling was attempted.

5.2.1 Data Structure

Let Y denote the plant outcome of interest, where in the present analysis the intended response variable was total nitrogen, denoted Total_N. Let

$$X^{(S)} = (X_1^{(S)}, X_2^{(S)}, X_3^{(S)}, X_4^{(S)})$$

represent the vector of single-band spectral predictors, and let

$$X^{(M)} = (X_1^{(M)}, X_2^{(M)}, \ldots, X_6^{(M)})$$

represent the vector of multiple-band spectral predictors. These predictors were extracted from the satellite-based spectral dataset and combined into a common analysis frame.

Prior to multivariate analysis, the predictor matrix was standardized so that each spectral variable had mean zero and unit variance. This ensured that all variables contributed comparably to the covariance and correlation structure underlying the dimension-reduction procedures.

5.2.2 Principal Component Analysis

Principal component analysis (PCA) was used as an exploratory tool to summarize the dominant modes of variation in the spectral predictors. Separate PCA models were fit to the single-band and multiple-band predictor sets. For the single-band data, the PCA was based on the standardized predictor matrix $X^{(S)}$, and for the multiple-band data, the PCA was based on $X^{(M)}$.

For each predictor group, scree plots, principal component standard deviations, loading matrices, and biplots were examined to assess the effective dimensionality of the data and to identify the variables contributing most strongly to the leading components. For exploratory purposes, the first two principal components from each group were retained as a compact summary of the major sources of spectral variation.

5.2.3 Factor Analysis

Because the scientific goal involved identifying interpretable latent constructs underlying the spectral measurements, exploratory factor analysis was also conducted separately for the single-band and multiple-band predictors. The factor analysis used principal-axis factoring, and the number of retained factors was guided by parallel analysis.

For both the single-band and multiple-band analyses, two latent factors were estimated using principal-axis factoring with oblimin rotation. The oblimin rotation was selected because correlation between latent factors was considered plausible in the spectral setting. The resulting latent factors were interpreted using the estimated loading matrices, and factor scores were obtained using regression-based scoring.

Specifically, the retained factor-score vectors were defined as

$$(LF_1^{(S)}, LF_2^{(S)})$$

for the single-band predictors and

$$(LF_1^{(M)}, LF_2^{(M)})$$

for the multiple-band predictors. These four latent variables were intended to provide a reduced-dimensional representation of the spectral information for later modeling.

5.2.4 Planned Regression Framework

The next planned stage of the analysis was to relate the latent spectral factors to plant nitrogen and carbon outcomes using regression models. In particular, the intended working model for nitrogen had the form

$$Y_i = \beta_0 + \beta_1 LF_{1i}^{(S)} + \beta_2 LF_{2i}^{(S)} + \beta_3 LF_{1i}^{(M)} + \beta_4 LF_{2i}^{(M)} + \varepsilon_i,$$

where

$$\begin{aligned} Y_i &= \text{nitrogen measurement for observation } i, \\ LF_{1i}^{(S)}, LF_{2i}^{(S)} &= \text{single-band latent factor scores for observation } i, \\ LF_{1i}^{(M)}, LF_{2i}^{(M)} &= \text{multiple-band latent factor scores for observation } i, \\ \varepsilon_i &= \text{random error term.} \end{aligned}$$

An analogous regression framework was intended for the carbon outcome. However, due to a data issue identified during the project, these regression models were not fit as part of the current analysis. As a result, the completed portion of this work focused on preprocessing, principal component analysis, factor extraction, factor-score construction, and interpretation of the latent spectral structure.

5.2.5 Methodological Rationale

The use of separate analyses for single-band and multiple-band predictors was motivated by the different informational roles of these spectral measures. PCA was used to understand the overall covariance structure and to visualize major directions of variability, while factor analysis was used to construct interpretable latent variables that could serve as predictors in a subsequent regression model. Although the final nitrogen and carbon regressions were not completed because of the data issue, the methodology established a reproducible framework for reducing the dimensionality of the spectral data and preparing it for later inferential modeling.

6 Findings

6.1 Findings: Split-Plot Analysis of Soil Aggregation

The main findings from the split-plot analysis came from the planned contrasts, which were designed to determine whether the effect of AMF differed between cover-crop and no-cover-crop conditions within Fusarium-infected and non-infected plots. These contrasts provided a more direct answer to the scientific questions than the individual model coefficients.

Under Fusarium-infected conditions, the evidence for a difference between cover-crop and no-cover-crop settings was limited. The comparison of AMF versus water showed a marginal difference, suggesting some possible indication that cover cropping may modify the AMF effect when Fusarium is present, but the evidence was not strong enough to conclude this definitively. The comparisons of AMF versus fungicide and AMF versus the average of fungicide and water were not statistically significant, indicating that under infected conditions there was no clear evidence that cover cropping substantially changed the relative performance of AMF on soil aggregation.

The pattern was more pronounced under non-Fusarium conditions. In non-infected plots, the AMF versus water contrast was statistically significant, indicating that the effect of AMF relative to the placebo treatment differed between cover-crop and no-cover-crop settings. The AMF versus fungicide contrast was also statistically significant, suggesting that cover cropping altered the relative effect of AMF compared with fungicide when Fusarium infection was absent. In addition, the comparison of AMF against the average of water and fungicide was statistically significant, reinforcing the conclusion that under non-infected conditions the AMF treatment behaved differently depending on whether cover cropping was present.

Taken together, these findings suggest that cover cropping had a clearer modifying effect on AMF-related treatment comparisons when Fusarium infection was absent than when it was present. In other words, the treatment differences involving AMF were more detectable in the non-infected setting, whereas under infected conditions the corresponding contrasts did not provide strong evidence of a meaningful difference between cover-crop and no-cover-crop management.

These findings should also be interpreted with some caution because multiple planned contrasts were tested. No formal multiple-comparison correction was applied in the present analysis, although such an adjustment would likely have been appropriate given the number of hypothesis tests conducted. As a result, the statistically significant findings, especially those near the conventional significance threshold, should be viewed as suggestive rather than definitive.

6.2 Findings: Spectral Analysis for Carbon–Nitrogen Monitoring

The spectral-data analysis was conducted as an exploratory step to determine whether the satellite-derived predictors exhibited interpretable low-dimensional structure that could later be related to plant carbon and nitrogen levels. Because the spectral variables were divided into single-band and multiple-band groups, the results are reported separately for each set.

For the multiple-band predictors, principal component analysis showed an extremely strong low-dimensional structure. The first principal component explained approximately 81.1% of the total variation, while the second explained about 18.9%, so that the first two components together accounted for essentially all of the variability in the multiple-band data. This indicates that the multiple-band predictors were highly collinear and could be summarized very effectively in a two-dimensional representation. Inspection of the loadings also showed that most of the multiple-band variables contributed strongly and in a similar direction to the first principal component, suggesting that this component captured a broad common spectral signal rather than a narrow feature driven by only one variable.

The single-band PCA also suggested meaningful dimensional reduction, although the structure was less concentrated than in the multiple-band case. The first principal component was dominant, but the remaining variation was more diffuse across later components compared with the multiple-band data. The biplot suggested that the single-band variables did not all behave identically, and that there was some differentiation in how bands contributed to the leading axes of variation. In practical terms, this means that the single-band predictors still contained redundancy, but their structure was somewhat less uniform than the multiple-band predictors.

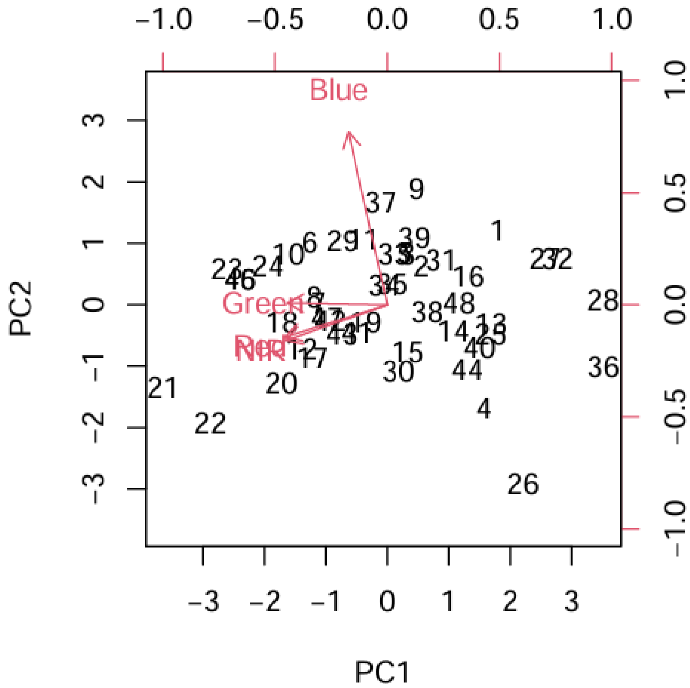

Figure 1: PCA biplot for the single-band spectral predictors. The figure illustrates the dominant low-dimensional structure in the single-band data and shows that the spectral bands do not all contribute identically to the leading principal components.

Parallel analysis was used to guide the factor-analytic part of the spectral investigation. For the single-band predictors, parallel analysis suggested retaining one factor. This indicates that the single-band measurements were largely driven by one dominant latent dimension. However, a two-factor model was still examined for exploratory purposes so that the structure could be compared more directly with the multiple-band case and to illustrate how latent variables would be constructed for downstream modeling.

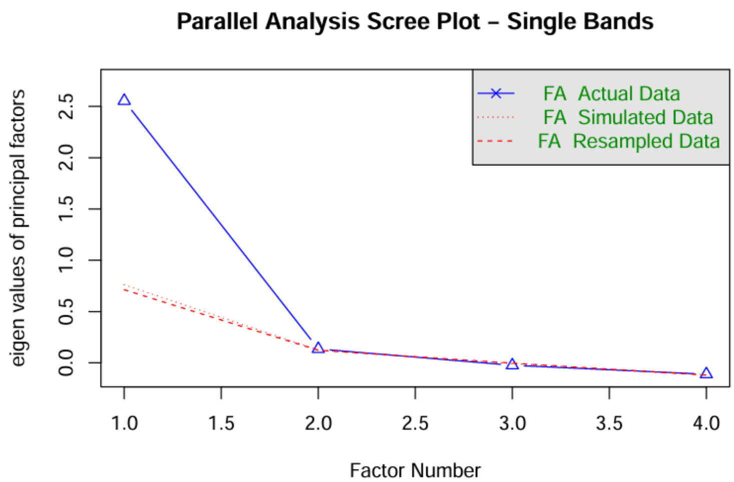

Figure 2: Parallel analysis scree plot for the single-band spectral predictors. The plot suggests retaining one factor, indicating that the single-band measurements are largely driven by a single dominant latent dimension.

For the multiple-band predictors, parallel analysis suggested retaining two factors. This result was more consistent with the PCA findings and supported the idea that the multiple-band measurements could be summarized by a small number of latent constructs. Because this analysis was intended to support later regression modeling, retaining two factors for the multiple-band data provided a useful compromise between dimensionality reduction and interpretability.

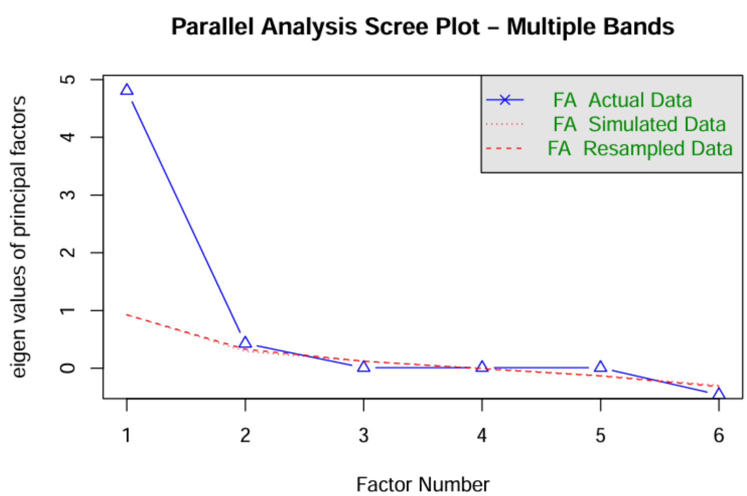

Figure 3: Parallel analysis scree plot for the multiple-band spectral predictors. The plot supports a two-factor solution, suggesting a richer latent structure than was observed for the single-band predictors.

The factor loadings for the single-band data indicated that one latent factor was driven primarily by the red and near-infrared bands, both of which loaded very strongly on the first factor. Green also loaded moderately on the first factor and to a lesser extent on the second factor, while blue contributed more weakly overall. This pattern suggests that the dominant latent structure in the single-band data was associated mainly with the red and near-infrared portion of the spectral signal, with the second factor providing only limited additional separation. Consistent with the parallel analysis result, the second factor in the single-band setting appeared relatively weak and should be interpreted cautiously.

The factor structure for the multiple-band predictors was much stronger. The first factor had very high loadings for SR, MTVI2, MSAVI, NDVI, and Red Edge NDVI, indicating that these variables were all capturing a common latent signal. The second factor was dominated by Green NDVI, with MTVI2 also showing a notable loading in the opposite direction. This suggests that the multiple-band data contained one broad common vegetation-response factor and a second factor that may reflect a more specific contrast involving the green-based index relative to the other derived vegetation measures.

Tthe factor solutions were useful for exploratory interpretation, but they should not be viewed as fully stable inferential results without further data cleaning or model refinement due to the previously mentioned data issues in the methods section.

7 Future Improvements

One area for future work concerns the spectral-data analysis. The client is currently working to resolve the data issues that prevented the regression stage from being completed. Once these issues are addressed, the factor analysis should be repeated to ensure that the latent structure is based on a cleaned and reliable dataset before proceeding to regression modeling. This will allow a more appropriate assessment of whether plant carbon and nitrogen levels can be meaningfully modeled using the satellite-derived spectral predictors.

A second area for future work involves the soil aggregation outcome used in the split-plot analysis. In the current study, soil aggregation was analyzed using the existing water-stable aggregate metric provided in the dataset. As a next step, I will develop a principal component analysis (PCA) approach for constructing a new soil aggregation measure from the underlying aggregation-related variables. This revised soil aggregation metric can then be incorporated into the split-plot modeling framework to reassess treatment and management effects using a data-driven summary of soil structure.